

VINEETH SAGAR H L

3rd Year B.Tech — ISE, BMS Institute of Technology

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Software Engineering and AI/ML student with hands-on experience building production-grade intelligent systems spanning agentic AI, RAG pipelines, and predictive models. Proven ability to design scalable, research-backed applications—from agricultural systems to cybersecurity threat detection—with a focus on clean code architecture and deployment-ready solutions

EDUCATION

B.Tech — Information Science and Engineering 3rd Year (2023–2027), CGPA :9.2 Pre University Course, PCMBs (2021–2023), 90%	BMSIT&M SIR M V PU college
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TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, TypeScript, SQL,Node.js, React.js. Frameworks: PyTorch, FastAPI, Flask, Pandas, NumPy, n8n. Tools: Git, Vercel, Docker (basic), Jupyter Domains: Machine Learning, RAG Systems, Agentic AI, Web Dev.

PROJECTS

Zero-Day Attack Detection Using TGNN and Explainable AI in Kubernetes Environments - Built an ML pipeline for zero-day attack detection using TGNN achieving 95.6% precision and 150ms detection latency. - Implemented temporal graph construction and trained multiple models (TGNN, LSTM-AE, DeepLog, Isolation Forest). - Integrated SHAP-based explainability with MITRE ATT&CK mapping for real-time behavioral anomaly detection
AgriGain — Agentic AI for Agricultural Waste Processing (part in bio Agri Eco-System) - Built an agent-driven AI system combining 400+ research papers, market data ingestion, and ROI computation. - Designed reasoning pipelines, modular orchestrators, reproducible tests, and deployment-ready architecture. Achievements: 3rd Prize (hack.io—KSSEM).
NetCDF RAG Pipeline — Scientific Retrieval-Augmented Generation - Developed a RAG system capable of interpreting scientific NetCDF (.nc) spatial–temporal datasets. - Implemented preprocessing, embeddings, vector search, and structured-output LLM reasoning.
E-Voting System with QR-Based Authentication and Real-Time Result Tracking - Developed a secure college-level E-voting platform with QR code-based voter verification and CSV-based voter list import for targeted student elections. - Implemented real-time result tracking dashboard with role-based access control (RBAC) for election administrators and voters. - Designed scalable registration and voting interface ensuring accuracy, transparency, and preventing duplicate voting with session management.

Plant Disease Detection System with Ensemble Deep Learning (part in bio Agri Eco-System)

- Built a full-stack AI application using ensemble ResNet50 models achieving 99.91% validation accuracy across 38 disease classes.
- Implemented ensemble system with ResNet50 and Channel-Augmented ResNet50 model with CUDA-accelerated inference for robust classification.
- Developed treatment recommendation engine with disease-specific solutions integrated into Next.js frontend and Flask backend.

Full-Stack Photography Portfolio Website (for friend)

- Built a full-stack photography website using Node.js/Express backend and React frontend with MySQL database and Sequelize ORM.
- Implemented JWT-authenticated admin panel with image upload , feedback management, and visitor tracking functionality.
- Designed responsive UI with frosted glass effects, animated gallery with cascade animations, mobile hamburger menu, and Intersection Observer-based scroll animations

ADDITIONAL INFORMATION

- **Languages:** English, Hindi, Kannada.
- **Achievements:** 3rd Prize (hack.io—KSSEM Hackathon), 2nd Prize(Build & Conquer Hackathon)